



UNIVERSITY OF THE FREE STATE

BLOEMFONTEIN CAMPUS

CSIS3744
Memorandum

**DEPARTMENT OF COMPUTER SCIENCE AND
INFORMATICS**

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**EXAMINATION: Main End-of-Year 2020
PAPER 1**

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TIME: 3 HOURS

MARKS: 100

INSTRUCTIONS: Answer ALL the questions. Write legibly. Number your questions clearly and correctly.

Question 1 [26 Marks]

HUBLOT systems using “hublot.co.za” domain deals in luxury watches in the country. The company has four devices, which are being used in their network. These devices are allocated host names and IPv4 or IPv6 addresses, which belong to the company’s domain. This is shown in the table below:

	Network device	Host name	IP address
(a)	Web server	Web	190.111.12.18
(b)	DNS server	DNS	192.168.18.21
(c)	Workstation 1	Station1	fe80:0000:0000:0000:e863:ac78:0b36:7e2d
(d)	Workstation 2	Station2	2002:ac56:00de:0000:a8dd:31ed:769a:63e7

With reference to the table above, answer the following questions taking care to indicate in your answers which network device or IP address you are referring to by means of the device number.

- (a) Specify the address type for each of the relevant IPv6 addresses. [2 Marks]
<<< 1) Local unicast address ✓ 2) Global unicast address. ✓ >>>
- (b) What are the default subnet masks for each of the IPv4 addresses? [2 Marks]
<<< 1) 255.255.0.0. ✓ 2) 255.255.255.0. ✓
- (c) Specify the classes that the IPv4 addresses belong to. [2 Marks]
<<< 1) Class B ✓ 2) Class C. ✓
- (d) For each of the IPv4 and IPv6 addresses, write down the part of the IP address that contains the network information. [4 Marks]
<<< 1) 190.111 ✓ 2) 192.168.18 ✓ 3) fe08:0000:0000:0000 ✓ 4) 2002:ac56:00de:0000 ✓
- (e) Workstation 1 sent a request to the web server to receive a web page. Write down the destination socket that was used for the request. [2 Marks]
<<< 190.111.12.18 ✓ :80 ✓ >>>
- (f) Two of the network devices’ IP addresses cannot be used on the Internet.
- i. Identify these two devices by writing down their fully qualified domain names. [4 Marks]
<<< 1) DNS ✓ hublot.co.za ✓ 2) Station1 ✓ hublot.co.za ✓ >>>
- ii. What protocol can be implemented to enable these devices to gain access to the Internet by means of a single shared IP address? [1 Mark]
<<< PAT in conjunction with DNAT ✓ >>>
- iii. Explain how this protocol functions. [3 Marks]
<<< This protocol replaces all outgoing packets’ source private IP addresses with the same single public source IP address. ✓ Incoming packets will therefore have the same public destination IP address which will be replaced by unique private destination IP addresses. ✓ This protocol uses the destination port addresses to determine which unique private destination IP address should be assigned to which incoming packet. ✓ >>>

- (g) Explain the difference between a recursive and iterative DNS query. [2 Marks]

<<< Recursive - A query that demands a resolution or the answer "It can't be found". ✓ Iterative - A query where the local server issues queries to other servers. ✓ >>>

- (h) Discuss the improvements OSPF (Open Shortest Path First) routing protocol made over RIP routing protocol. [4 Marks]

<<< OSPF can be used on interior or border routers. ✓
 OSFP imposes no hop limits. ✓
 OSFP uses a more complex algorithm that involves more routing metrics when calculating the best path instead of just using the hop count. ✓
 It calculates the most efficient path rather than the shortest direct path. ✓
 It maintains a database of other routers' links. ✓
 It has a faster convergence time. ✓ >>>

Question 2 [20 Marks]

Study the following header and payload information of an intercepted network packet. The information has already been translated into readable text format.

Bit 0-7		Bit 8-15			Bit 16-23			Bit 24-31		
4	20 bytes	0			40 bytes					
0					1	0	0			
60		TCP			Correct					
172.13.14.15										
192.168.1.1										
443					49153					
3043958669										
937013559										
20 bytes		0	1	0	0	1	0	3200 bytes		
Correct										

- (a) Which Transport layer protocol was used during the transmission of the packet? [1 Mark]
 <<< TCP ✓ >>>
- (b) Which Application layer protocol was used during the transmission of the packet? [1 Mark]
 <<< HTTPS ✓ (Port # 443 in the TCP header). >>>
- (c) How many hops can the packet still be forwarded before it will be discarded? [1 Mark]
 <<< 60 ✓ (TTL field in the IPv4 header). >>>
- (d) What is the total size of the packet? [1 Mark]

<<< 40 bytes ✓ (Total length field in the IPv4 header). >>>

- (e) Explain the purpose of the “3200 bytes” value in the header. [2 Marks]

<<< The value of 3200 bytes is the sliding window size ✓ which indicates that the server (source) can accept 3200 more bytes of data from the client (destination) even when this segment remains unacknowledged. ✓ >>>

- (f) Provide the source and destination socket addresses for the transmission. [4 Marks]

<<< Source: 172.13.14.15 ✓:443 ✓ (Source IP and Port #)

Destination: 192.168.1.1 ✓:49153 ✓ (Destination IP and Port #) >>>

- (g) The packet illustrated above was the second packet of a three-way handshake transmitted between a workstation and a webserver on the Internet.

- i. Please illustrate how the third packet of the three-way handshake will look by specifying its sequence number, acknowledgment number, source socket, destination socket and transport layer flags that will be set. [5 Marks]

<<< Sequence #: 937013559 ✓

Acknowledgement #: 3043958670 ✓

Source socket: 192.168.1.1:49153 ✓

Destination socket: 172.13.14.15:443 ✓

Flags: ACK ✓ >>>

- ii. Explain three characteristics about this protocol that distinguishes it from UDP. [3 Marks]

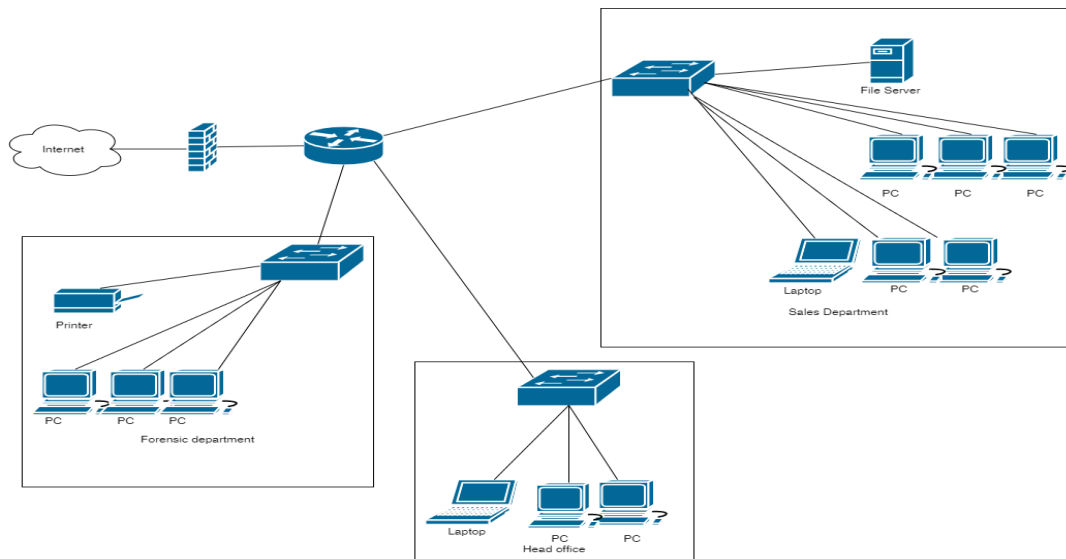
<<< TCP is connection-oriented; ✓ uses sequencing and checksums; ✓ and provides flow control. ✓ >>>

- iii. What is the difference between dynamic ARP table entries and static ARP table entries? [2 Marks]

<<< Dynamic ARP table entries are created when a client makes an ARP request, ✓ whereas static ARP table entries are entered manually using the ARP utility. ✓ >>>

Question 3 [34 Marks]

CHADEMA financial services has forensic department, sales department and a head office. See an illustration of this network in the diagram below.



- (a) CHADEMA financial services realized some security threats, although they have a firewall. Identify two security threats that this network could suffer. [2 Marks]
 <<< The router and firewall may not prevent all security threats. Therefore, threats like man-in-the middle attacks (DNS, HTTPS, IP, ARP spoofing, SSL hijacking), computer virus & worm, DOS/DDOS attack, phishing, etc. ✓✓>>>
- (b) Discuss the mitigation procedures for the identified threats in Question (a) above. [2 Marks]
 <<< Implement proxy server and DMZ are just of the answers expected from candidates ✓✓>>>
- (c) Identify the type of router that connects CHADEMA financial services to the Internet. [1 Mark]
 <<< Border router. ✓>>>
- (d) On what level of the OSI 7 layer reference model does a router operate? [1 Mark]
 <<< Network layer. ✓>>>
- (e) Discuss the IS-IS link state routing protocol. [4 Marks]
 <<< IS-IS uses a best-path algorithm similar to OSPF's. ✓ IS-IS is designed for use on interior routers only (unlike OSPF). ✓ IS-IS is not handcuffed to IPv4 (like OSPF) so it's easy to adapt to IPv6. ✓ Service providers generally prefer IS-IS because it's more scalable than OSPF. ✓>>>
- (f) A PC at the sales department is unable to access the network. When you check the LED lights near the PC's network port, you discover the lights are not lit.
- Which layer of the OSI 7 layer reference model are you using to troubleshoot this problem? [1 Mark]
 <<< Physical layer ✓>>>
 - At which two layers does the network adapter work? [2 Marks]
 <<< Data link layer ✓ and physical layer ✓>>>
 - Why is it important to use a structured cabling standard when installing and managing cabling systems? [2 Marks]
 <<< The standard suggests how networking media can best be installed to maximize performance ✓ and minimize upkeep. ✓>>>

- (g) A PC at the head office with the IP address 192.168.1.1 sends a request for a webpage to an Internet webserver with the IP address 172.13.14.15. Thereafter the webserver responds to the PC's request. Note that CHADEMA financial services has an ADSL Internet connection with their router's internal IP address being 192.168.1.254 and the router's public IP address being 72.131.141.151. Illustrate how NAT in combination with PAT is implemented on the router by completing the table below. Specify the source and destination socket addresses as the request and response travels across the Internet. [8 Marks]

	Request/response	Source socket	Destination socket
(i)	The request travels from the PC to the router	192.168.1.1:49153✓	172.13.14.15:80✓
(ii)	The request travels from the router to the webserver.	72.131.141.151:49153✓	172.13.14.15:80✓
(iii)	The response travels from the webserver to the router.	172.13.14.15:80✓	72.131.141.151:49153✓
(iv)	The response travels from the router to the PC.	172.13.14.15:80✓	192.168.1.1:49153✓

- (h) Explain in detail the role that the DNS protocol played during the transmission of the request with regard to local, top level and root name servers. [4 Marks]
 <<< DNS queries the local name server for the IP address of the specified web server (www.chademafinancials.co.za) ✓ If the local name server cannot provide the required information, DNS will query the root name server for the IP address of the top level .co.za name server. ✓ DNS will then query the .co.za name server for the IP address of the authoritative chademafinancials name server. ✓ Finally, DNS will query the chademafinancials name server for the IP address of the web server. ✓
- (i) Some of the employees of CHADEMA financial services connect to a LAN from a remote location using the Point-to-Point Protocol (PPP).
- What server is necessary for this connection? [1 Mark]
 <<< Remote access server ✓>>>
 - Why is Point-to-Point Protocol (PPP) a better protocol than Serial Line Internet Protocol (SLIP)? [5 Marks]
 <<< Any 5 points earns full marks: SLIP can carry only IP packets, whereas PPP can carry many different types of Network layer packets. ✓ When using SLIP, you typically must specify the IP addresses for both your client and for your server in your dial-up networking profile. PPP, on the other hand, can automatically obtain this information as it connects to the server. ✓ PPP also performs error correction and data compression, but SLIP does not. ✓ In addition, SLIP does not support data encryption, which makes it less secure than PPP. ✓ SLIP only works on serial connections. ✓ SLIP supports only asynchronous data transmission, and PPP supports both asynchronous and synchronous transmission. ✓ >>>

- iii. Which secured tunnelling protocol might be able to cross firewalls where IPsec is blocked? [1 Mark]
<<<OpenVPN. ✓>>>

Question 4 [10 Marks]

When running a scan on your computer, you find that a session has been established with a host at the address 208.85.40.44:80.

- (a) Which Application layer protocol is in use for this session? [1 Mark]
<<< Port 80 indicates this is an HTTP session. ✓>>>
- (b) What command-line utility might you use to determine which computer is the host? [2 Marks]
<<< nslookup can identify the domain name of the host at that IP address. ✓✓>>>
- (c) Explain two reasons why wireless networks experience a greater reduction in throughput as compared to wired networks. [2 Marks]
<<< Wireless networks experience a high number of collisions ✓ and require greater overhead on each transmission, ✓ resulting in a significant reduction in throughput compared to wired networks.>>>
- (d) An attacker configures a VLAN frame with two tags instead of just one. The first tag directs the frame to the authorized VLAN. After the frame enters the first VLAN, the switch appropriately removes the tag, then discovers the next tag, and sends the frame along to a protected VLAN, which the attacker is not authorized to access. What kind of attack is this? [2 Marks]
<<< Either VLAN hopping or double tagging. ✓✓>>>
- (e) What kinds of issues might indicate a misconfigured ACL? [3 Marks]
<<< Connectivity ✓ and performance issues ✓ between two hosts in which some applications or ports can make the connection while others can't ✓.>>>

Question 5 [10 Marks]

The Imperial house makes use of the public network address, 195.234.0.0.

- (a) To what network class does the above mentioned network address belong? [1 Mark]
<<< class C ✓>>>
- (b) The Imperial house wants to divide their network into two subnets.
i. Calculate the new subnet mask for the proposed sub networks. [3 Marks]

<<< Default subnet mask for class C network is 255.255.255.0.
Binary value of subnet mask is 11111111.11111111. 11111111.00000000

$2^n - 2 = 2$ $n = 2$ ✓
Change 2 bits from 0 to 1 for the last octet.

11111111.11111111.11111111.11000000 ✓

Value of 11000000 $\rightarrow 2^7 + 2^6 = 128 + 64 = 192$

The new subnet mask is 255.255.255.192 ✓

OR

Default subnet mask for class C network is 255.255.255.0.

Binary value of subnet mask is 11111111.11111111. 11111111.00000000

$2^n = 2$ $n = 1$ ✓

Change 1 bit from 0 to 1 for the last octet.

11111111.11111111.11111111.10000000 ✓

Value of 11000000 $\rightarrow 2^7 = 128$

The new subnet mask is 255.255.255.128 ✓

>>>

- ii. How many usable addresses are available per subnet? [2 Mark]

$\lll (2^6 - 2) = 62$ ✓

OR

$(2^7 - 2) = 126$ ✓ >>>

- iii. Complete the table below by supplying each subnet's network address, broadcast address and usable host address range: [4 Marks]

	Subnet	Network address	Broadcast address	Usable host address range
(i)	1	195.234.0.64 ½	195.234.0.127 ½	195.234.0.65 ½ to 195.234.0.126 ½
(ii)	2	195.234.0.128 ½	195.234.0.191 ½	195.234.0.129 ½ to 195.234.0.190 ½

OR

	Subnet	Network Address	Broadcast Address	Usable Host Address Range
(i)	1	195.234.0.0 ½	195.234.0.127 ½	195.234.0.1 ½ to 195.234.0.126 ½
(ii)	2	195.234.0.128 ½	195.234.0.255 ½	195.234.0.129 ½ to 195.234.0.254 ½

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